

Influencing customer's purchase intentions through firm participation in online consumer communities

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Abstract Today, many firms are striving to enhance customers' purchase intentions by leveraging online consumer communities but a better understanding of the underlying phenomena is needed to guide their efforts most effectively. In this study we analyse how three hotel firms engage potential customers through FlyerTalk, an online consumer community of frequent travellers (FlyerTalk.com). Our results suggest that for some firms, the frequency and perceived value of their participation are two important factors that can potentially be managed toward positive purchasing intention effects.

Keywords Forums · Online communities · Boundary spanning · Web 2.0 · Social marketing

JEL Classification D71 · D83 · M30 · O33

Introduction

It is generally recognized that social networking can be an extraordinarily powerful marketing tool that organizations of all kinds must strive to leverage effectively, but a better understanding of the underlying phenomena is still needed

to guide their efforts and ensure optimal results (Adjei et al. 2010). Social networking sites of particular interest to many organizations are *online consumer communities*—interactive sites where consumers discuss products, services, and special interests with other consumers and with company representatives (Puri 2007; Arnone et al. 2009; You et al. 2012). Research has shown that such communities can successfully influence consumer sentiment toward enhanced purchase intentions thereby increasing sales (Adjei et al. 2010) and firms now commonly invest in them seeking to do so. There are two main types of online consumer communities, (1) firm-hosted/controlled community sites that focus specifically on one company's products and services, e.g. Toyota's official owners' community¹ and (2) third-party-hosted community sites that are not controlled by the firm or firms whose products and services are being discussed, e.g. unofficial Toyota community sites.² In this paper, we focus on the third-party-hosted sites (those *not* controlled by their associated firms) because the effectiveness of firm-controlled sites appears to be limited in some contexts by the perception that interactions unfavourable to the firm may be suppressed through content filtering (Walden 2000; Arnone et al. 2009). Consumers perceive product and service reviews found in third-party-hosted online communities as more credible and valuable than those offered on firm-controlled sites, particularly for experience products such as movies, vacations, etc. (Bae and Lee 2011).

Firms may also benefit from these types of consumer communities and the boundary spanning opportunities they provide as bridges linking internal (human) networks to those of consumers. According to Aldrich and Hecker, "two classes of functions are performed by boundary roles: information processing and external representation" (1977, p. 218). Online consumer

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¹ <http://www.toyota.com/owners/web/pages/community> last accessed on October 02, 2013

² <http://www.toyotafans.net/> last accessed on October 02, 2013

communities can facilitate information gathering and processing by providing unbiased reviews of products/services and the desires of a wide demographic of consumers (Dellarocas 2003; Amis 2007; Xu et al. 2011; You et al. 2012). Scanning forums for this information allows firms to improve their current offerings, better tailor future designs with less risk, and glean insights into competitors' products (Aldrich and Hecker 1977; Xu et al. 2011; Dellarocas 2003; You et al. 2012).

Participation in consumer communities can also extend firms' external representation efforts as firm representatives may harness these sites to improve customers' image of the firm and its products and services (Aldrich and Hecker 1977; Bettencourt et al. 2005). For example, some research suggests that firms may derive value, such as increased brand awareness and loyalty, from building relationships with existing and potential consumers through authentic engagement in online consumer communities (Amis 2007; You et al. 2012). However, participation in online consumer communities is not without its risks. Posts made by company representatives are likely to be viewed by consumers as official statements made by the firm whether they are meant to be or not, and social missteps may be seen, and promulgated, by thousands of customers.

Although there is real industry interest in employing social media for such boundary spanning activities, few studies offer insight as to how firms should manage their participation in online consumer communities to best leverage the boundary spanning opportunities this media provides. In this study we address this gap, extending the boundary spanning literature with respect to customer-facing external representation, and advancing the application of social media for business value through an empirical analysis of three firms' participation in an online consumer community. The firms' external representation agents each bridge the consumer network to that of their firm, seeking to influence purchase intention in the process, but in this third-party-hosted context their locus of control is limited to the corpus of postings/responses they proffer for the community's consumption. Clearly, their success is dependent on the nature of their contributions and, seeking insight to help guide their efforts, we formulate our fundamental research question as follows:

With respect to a firm's boundary spanning external representation by agents participating in third-party-hosted online consumer communities, what are key, potentially manageable parameters of posting activity that can be shown to influence purchasing intention?

To address this research question credibly, it was necessary to study engagement by real world firms in an established, substantial third-party-hosted online consumer community and we settled on FlyerTalk for our context. FlyerTalk was established in 1998 as an Internet discussion forum for air

travellers to seek and provide advice on frequent flyer programs and travel-related issues and has grown to become one of the most successful examples of a third-party-hosted online consumer community. It afforded us the opportunity to identify a set of prominent competing firms engaging consumers there and to explore how their efforts affect their relative success in influencing purchasing intention.

The remainder of the paper is organized as five sections. In the first of these, we provide a brief background on FlyerTalk and on the three hotel brands included in this study: Marriott, Hilton and Starwood. In the next section, we turn to the boundary spanning and marketing literatures to guide our development of a model of firm involvement in online consumer communities. The subsequent section explains our methodological approach to testing our model via a survey of FlyerTalk forum members and we also present some basic analytic results. The section following that provides a further analysis, results, and discussion of the study's findings. The final section, Conclusion, addresses limitations, suggests future research, and summarizes the study's implications for today's firms.

Background

Herein we investigate the activities of three hotel chains: Hilton, Marriott, and Starwood, as they compete for visitor engagement in the online consumer community FlyerTalk. FlyerTalk is a third-party-hosted, self-moderated community of frequent business travellers who share a common interest in maximizing the value of their travel arrangements. Our selection of FlyerTalk was based upon its longevity – 16 years, its popularity – 27,000 to 52,000 daily unique visitors and 20 million page views per month, and the number and frequency of posts – over 20 million posts to date.

Three major competitors from the hotel industry were chosen due the interest in these hotels and their rewards programs amongst a wide range of FlyerTalk members. Hilton Worldwide is a privately held international chain of hotels (DoubleTree, Embassy Suites, Hampton Inn, etc.) with over 4,000 properties across 90 countries. Their HHonorsRepresentative began promoting Hilton's reward program, HHonors, in FlyerTalk on June 25th, 1999. Marriott is an international company with more than 3,000 properties (Ritz-Carlton, Renaissance, etc.), an annual revenue topping \$13 billion and over 150,000 employees. Marriott's involvement in the FlyerTalk forums, via the Marriott Concierge, began in late 2003. Starwood Hotels and Resorts Worldwide (Sheraton, Westin, etc.) owns over 1,100 properties and has more than 171,000 employees. The company's involvement in FlyerTalk began on Nov 22, 2000 with the introduction of a single employee representative, William Sanders, known on the boards as the Starwood

Lurker. Together these three firms, of similar size and profile, dominate the industry and compete directly across their constituent brands, targeting the same traveller segments in advertising and, notably for our purposes, their affinity programs.

Given the service-based nature of the hotel industry, it is not surprising that research shows customer satisfaction and image perceptions to be correlated to loyalty and, by extension, to firm success (Kandampully and Suhartanto 2000), so we refer to the marketing literature, specifically in the areas of customer service and boundary spanning between firms and their customers, to guide our investigation into how differences in levels of online community involvement may explain different outcomes.

Theoretical development

Over the last five decades a rich literature has developed around boundary spanning—communication that crosses an organization's boundary with its environment (Aldrich and Hecker 1977; Leifer and Huber 1977; Lefier and Delbecq 1978; Katz and Kahn 1978). Research on boundary spanning covers a wide array of possible linkages between organizational members and non-members to include information exchange between research teams or organizational units, companies and their suppliers, organizations and the public, firms and their customers (Aldrich and Hecker 1977; Springston and Leichty 1994; Manev and Stevenson 2001; Marrone et al. 2007). Many of these studies focus on boundary spanning individuals—those within the organization with the knowledge, skills and expertise to effectively communicate with those outside the organization—or boundary spanning activities (Leifer and Huber 1977; Tushman and Scanlan 1981a, b; Dollinger 1984; Richter et al. 2006).

Customer-facing or customer-oriented boundary spanning is one subset of that research that is of particular relevance to the current study because a firm's engagement in third-party-hosted online communities like FlyerTalk is clearly aimed at its customer base. Such studies investigate the influence of employee boundary spanning activities on organizational operational measures such as employee turnover, financial performance and customer behavioural intention. Their models incorporate factors such as customers' identification with the company, product utilization behavior, firm image, service quality perceptions and satisfaction (Dollinger 1984; Ahearne et al. 2005; Bettencourt et al. 2005).

In order to investigate how firms should participate in online consumer communities to maximize the benefits of their external representation efforts, we searched the literature for dependent variables of firm benefit appropriate for our focal firms and context. Research suggests that boundary spanning activities may positively impact firm performance

either directly or indirectly (Dollinger 1984; Rosenkopf and Nerkar 2001; Ahearne et al. 2005). For example, Dollinger revealed a direct, positive relationship between business owners' boundary spanning activities and financial performance (1984). The performance measures for Dollinger's study of very small firms' net retained earnings, current sales and gross economic benefits accrued to ownership—are not applicable to our study of three large hotel firms as the number of intervening factors between firm participation in online forums and financial performance is too great to isolate the impacts these boundary spanning activities. Indeed, many organizational performance measures are equally problematic. However, Ahearne et al. (2005) found that consumers' positive perceptions of a firm's external image and characteristics as well as characteristics of the firm's boundary spanning agents can increase consumers' identification with the firm and ultimately their product use behaviors. The marketing literature also suggests a number of performance measures influenced by consumers' perceptions of boundary spanning activities. For example, numerous studies assess firm performance as consumers' purchasing intentions or as customer loyalty and retention (Zeithaml 2000; Parasuraman et al. 1988; Zeithaml et al. 1996; McKinney et al. 2002; Heere et al. 2011). Drawing from the intersection of these two literatures we chose our measure of firm performance to be consumers' purchasing intentions, operationalized in our context as hotel choice preference, or more specifically, the preference to choose a particular hotel brand over others.

But which characteristics of customer-facing boundary spanning activities influence business outcomes? The boundary spanning literature offers some suggestions. Katz and Kahn (1978) suggest that the quantity of time spent boundary spanning (i.e. intensity) and the number of contacts reached by boundary spanning (i.e. extensity) both influence the effectiveness of boundary spanning activities. Likewise Dollinger (1984, pp. 354) concludes that “higher levels of boundary spanning are associated with higher levels of [organizational] financial performance”. Although these studies examine different aspects of the phenomenon, taken together they suggest that more boundary spanning activity results in better performance. Indeed marketing researchers have suggested that higher levels of firm participation in “brand communities” (online communities focused exclusively on a particular brand) may lead to increases in loyalty and repurchasing intentions for favoured brands (McAlexander et al. 2003; Thompson and Sinha 2008; Heere et al. 2011). That boundary spanning through branded communities is thought to be capable of contributing to greater purchasing intention leads us to speculate that, by extension, the same may hold for firms' participation in brand-independent, third-party-hosted online communities like FlyerTalk despite the contextual dissimilarities such as the absence of brand firm control over members' postings and the consequent need to conform to the brand-heterogeneous community's

expectations and standards of conduct. In this context, firm participation would be reified as posting messages in the forums—the greater posting frequency, the higher the level of firm participation and the greater the effect on purchasing intention. Indeed, communication frequency (i.e. the quantity of communication over a period of time), has been recognized as a key factor in traditional advertising models—greater communication frequency results in increased product awareness and purchasing intentions among consumers (Ray and Sawyer 1971; Tellis 1988) and so it follows, analogously, that this form of communication frequency may serve to help gauge a firm's boundary spanning engagement and the level of consequent purchasing intention benefit it engenders. Thus we propose the following:

H1: For firms posting in third-party-hosted online communities, greater levels of post frequency, as perceived by consumers, effects greater levels of consumers' purchasing intentions.

In addition to communication frequency, the marketing literature focusing on the area of customer service experience in general, also suggests other variables related to external representation that may impact behaviour/purchasing intention, including sacrifice (the monetary or non-monetary costs consumers incur when acquiring or using a service), customer satisfaction, and both the perceived quality and perceived value of customer service. Researchers offer several models of the relationships between these variables (Zeithaml 2000; Parasuraman et al. 1988; Zeithaml et al. 1996; McKinney et al. 2002; Lee and Lee 2009). Cronin et al. (2000) summarize the accumulated literature at point of publication, identifying three prominent models and adding a forth as follows:

1. a Value Model in which behavioral intentions are influenced directly only through value which is influenced by satisfaction, along with quality and sacrifice,
2. a Satisfaction Model in which behavioral intentions are influenced directly only through satisfaction which is influenced by value and by quality, with quality (along with sacrifice) also influencing value so that it has both a direct and indirect (through value) influence on satisfaction,
3. an Indirect Model which replicated The Satisfaction Model (2) but removes the direct influence of quality on satisfaction and restores the direct effect of value on behavioral intention, like (1), so that both value and satisfaction directly influence intention, and
4. their own model which refines (3) by restoring the direct influence of quality on satisfaction and more significantly, includes quality as directly affecting behavioral intention, along with value and satisfaction, so that quality influences intention both directly and indirectly, through

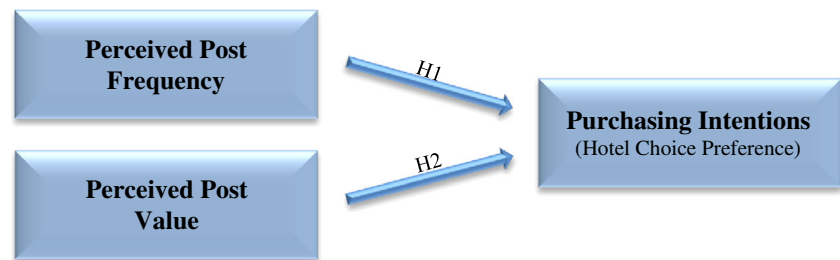
satisfaction. They found support for their model though their results suggest that service quality plays a much greater role in perceived service value than does sacrifice.

Since forums like FlyerTalk can be viewed as a form of customer service, we draw from the existing literature in that field to identify key factors at play. But at the same time the unprecedented aspects of an innovative new paradigm like this require an exploratory approach that postulates a very basic initial model as a foundation, to be refined as our understanding evolves. For example, the posting activity itself, coming into existence as part of the Web 2.0 revolution, enables members of third-party-hosted online communities, like FlyerTalk, to derive customer service benefits *indirectly*, simply by observing fellow members' interactions with a firm's representatives. This form of engagement has only become possible to any meaningful extent, with the advent of this new customer service paradigm, so while the existing literature guides us, we must find the right application of theory for this context and we begin by drawing on core concepts.

From traditional customer service research, we know that perceived service value plays a key role in influencing behavioral intention and value is in turn influenced by perceived service quality. Other factors such as sacrifice and satisfaction have been identified too, but in applying this framework to our context, we simplify, to develop a starting foundation, and we analogize existing knowledge as thoughtfully as we can to the case at hand. Thus, drawing from the literature's identification of perceived service value as a key factor, we suggest that a primary factor of posting activity to which community members will be attuned is the usefulness, or the perceived value, of the postings themselves which, analogously to the customer service models, is influenced by and thus encompasses the perceived quality of the informational content contained therein. Perceived posting value thus seems likely to be a material factor, based on what we know from traditional literature and at the same time provides a simple, generalized measure appropriate to an exploratory study of this type. If the hypothesis holds, then future research can build on it to tease out the interplay of value components as well as the roles of other factors drawn from traditional customer service literature and elsewhere. For this initial study however, we seek to establish a baseline understanding and thus propose the following:

H2: For firms posting in third-party-hosted online communities, greater levels of post value, as perceived by consumers, effects greater levels of consumers' purchasing intentions.

Figure 1 below summarizes the resulting theoretical model to be used as a basis for the study.

Fig. 1 Theoretical model

Methodology and results

To test our hypotheses we collected primary data from FlyerTalk forum members via a short survey posted to a brand-independent subsection of the forum. This provided us with access to a large, active and highly diverse respondent pool and reduced the influence of biases one would expect to find in brand-specific forums. By posting to a public forum, we gained access to experienced, valuable, and impartial respondents that would otherwise prove very difficult to get. We incentivized survey participation by offering thirty \$10 Amazon gift cards and by minimizing the length of the survey.

The survey was designed to collect a small set of demographic and descriptive data along with measures of the explanatory variables—perceived frequency and value, and the dependent variable—intention to stay. Since the proposed model is based on basic, well-recognized antecedents suggested by the marketing and boundary spanning literature, it was not necessary to develop complicated, multi-item measures of constructs as idealized conceptual abstractions. (This was particularly fortunate since brevity was deemed key to attaining reasonable completion rates in this type of environment where users tend to be impatient). The unambiguous nature of “perceived frequency” allows for capture by a single item and its quantifiability makes it easy to assess the validity of users’ perceptions objectively, as we do below. The appropriateness of single-item measures for basic, self-report measures such as these is well-established. For example, Bergkvist and Rossiter (2007) found single item measures to provide a predictive ability similar to that of multi-item measures for marketing constructs such as attitude toward brand. Perceived value and hotel choice preference are arguably similarly well understood though they are less objectively quantifiable for validating perceptions. Clearly, the term “value” can connote more abstraction and complexity, but we conceptualize “perceived value” in our context at a generalized level, seeking in this initial exploratory study to identify whether, in the broadest sense, it plays a material role at all. Assuming the generalized construct’s impact is established, the relative effects of component dimensions would be explored in further research with more complex,

multi-item measures. But for our purposes, a single item is sufficient, and ideal even, to measure a generalized, cogent perception of value, encompassing the breadth of individual subjects’ perspectives, amalgamated across the full subject pool.

The FlyerTalk moderator granted us permission to post our survey to a forum for one week’s time during which 182 respondents completed the survey. Of these, 39 respondents belonged to reward programs that were not a part of our study and their responses were removed. Four responses with missing data were also removed leaving 139 usable responses. Summary statistics of respondent demographics are included in Table 1.

As seen in Table 1, the respondents were fairly evenly distributed across the three reward programs we were studying, providing a sound basis for comparing results among them. The business/personal split was also roughly even. The vast majority of respondents were located in North America, and the average period active on the site was nearly 3 years, with a notably dichotomous distribution showing more newer and older respondents than mid-range ones. Also, the vast majority (89 %) of respondents had been active in the community for at least a year with a large portion (40 %) being active for at least 5 years. So in general, respondents had enough experience to have developed well-informed perceptions, even for the service with the lowest posting rate (40/year). To ensure that the respondents are reflective of the general population of FlyerTalk users, we posted the survey in the brand independent, Travelbuzz sub-forum, instead of a hotel specific sub-forum. Indeed when dividing the responses by reward program, the descriptive statistics showed no significant differences between the three groups.

Survey respondents were directed to different survey rating item sets customized to their preferred hotel reward program. For example, if a subject selected Hilton HHonors as their preferred reward program, he or she was given a set of rating items pertaining to the forum participation of the HHonors Representative. Similarly, those respondents that selected Marriott or Starwood’s reward programs were directed to rating items pertaining to the Marriott Concierge or the Starwood Lurker, respectively. Each respondent’s rating item set was the same in

Table 1 Descriptive statistics of survey respondents

Item	Descriptive statistics	Response	N	%
Primary purpose for travel	<i>N</i> =139	Business	77	55 %
		Personal	62	45 %
Place of residence	<i>N</i> =139	North America	128	92 %
		All Others	11	8 %
Years active on FlyerTalk	<i>N</i> =139 Mean=2.87 SD=1.06	Less than 1 Year	15	11 %
		1 to 3 years	44	32 %
		3 to 5 years	24	17 %
		5 years or more	56	40 %
Preferred hotel reward program	<i>N</i> =139	Hilton HHonors	54	39 %
		Marriott Rewards	36	26 %
		Starwood Preferred Guest	49	35 %

terms of content but reflected the hotel representative specific to the respondent's selected reward program:

- The [hotel reward program] representative posts frequently on FlyerTalk
- Posts made by the [hotel reward program] representative on FlyerTalk are highly valuable to consumers
- I am more likely to stay at a property of [the hotel] because of the [hotel reward program] representatives' posts on FlyerTalk

(Note that the phrase "...because of the [hotel reward program] representatives' posts on FlyerTalk" was deliberately and necessarily incorporated into the third item in order to isolate the effects of the hotel representatives' actions by excluding any number of unrelated company efforts that might impact intent to stay (eg. prior experiences with the brand or even posts made by other members of Flyertalk). Its use here does not confound the study's goals of determining whether the two factors post frequency and post value, *specifically*, influence consumers' intentions to stay and to gauge that influence, if detected.)

For each item, respondents were asked to indicate their level of agreement on a scale of 1–5 (1=strongly disagree, 5=strongly agree) in relation to our hypotheses. A summary of descriptive statistics for these three questions is presented in Table 2 below, grouped by hotel program. Included in the table are the p-values for independent (unpaired) sample t-tests of differences in the means across the scores for the three representatives with significance noted in the differences between Starwood Lurker and both Marriott Concierge and HHonorsRepresentative for each survey item, while the latter two were not significantly distinguishable from each other on any one. Finally, we collected additional data not included in this study for further research.

To assess whether forum members' perceptions of post frequency reflected reality we also collected secondary data

regarding the lifetime posting counts for each hotel representative as provided by FlyerTalk. Since joining in 2000, the Starwood Lurker has averaged 6.02 posts per day. In comparison, the Marriott Concierge and HHonorsRepresentative average only 0.29 and 0.11 posts per day respectively. Thus forum members' perceptions of post frequency match the actual forum activity of the three hotel representatives.

By contrast, there is no corresponding objective measure of post value, so members' perceptions cannot be so definitively validated, but subjective support was found, anecdotally, through targeted examination/comparison of actual post contents. Searching each of the three hotels' reward program forums using keyword combinations such as "Starwood Lurker and Marriott Concierge and value," and "Hilton HHonorsRepresentative and Marriott Concierge and value," revealed several references to the perceived disparities in the value of hotel representatives posts including:

- "THX! Boy, do I wish that all my loyalty programs had a "lurker" on site that was as responsive and helpful as you [the Starwood Lurker]!! One more reason I'm switching to SPG after 2 years platinum with Marriott & Diamond Hilton."
- "Is MP reaching out to the programs or just letting the programs figure it out? The Starwood Lurker's are very useful while some of the other reps seem, um, less useful..."
- "We have our own SPG "angel" here at flyertalk, Starwood Lurker (William). William is super responsive in not only answering questions about the program here on the forum, but also via email and has solved several issues I've had."

Notably, the keyword searches did not return comparisons that favored the HHonorsRepresentative or the Marriott Concierge nor any accolades for these representatives. These examples corroborate survey response summaries of

Table 2 Descriptive statistics summary

Survey item	Descriptive statistics	Hotel program	Mean	Std Dev	p-values for t-tests of Differences in means	
					Marriott	Hilton
The [hotel reward program representative] posts frequently on FlyerTalk	N=139	Starwood	4.14	0.89	7.09E-10***	1.32E-12***
	Mean=3.22	Marriott	2.86	0.76	–	0.21193
	SD=1.14	Hilton	2.61	1.02	–	–
Posts made by The [hotel reward program representative] on FlyerTalk are highly valuable to consumers	N=139	Starwood	4.31	0.80	1.03E-07***	3.35E-08***
	Mean=3.63	Marriott	3.28	0.81	–	0.85261
	SD=1.00	Hilton	3.24	0.99	–	–
I am more likely to stay at a property of [the hotel] because of the [hotel reward program representatives'] posts on FlyerTalk	N=139	Starwood	3.20	1.0	6.39E-02*	5.80E-04***
	Mean=2.81	Marriott	2.75	1.23	–	0.27215
	SD=1.12	Hilton	2.48	1.06	–	–

*** indicates significance at $p < 0.01$, ** indicates significance at $p < 0.05$; * indicates significance at $p < 0.10$

perceived post value. Additionally, examining statistics provided by FlyerTalk revealed that the Starwood Lurker generates 38 % (average 29 responses) more responses per thread than does the HHonorsRepresentative and 107 % more responses than the Marriott Concierge. Although response statistics do not assess post value they do illustrate that the Starwood Lurker's posts are more effective in engaging forum members than are posts made by the other two hotels' representatives.

Analysis and discussion

We tested our two hypotheses with linear regression. In an exploratory study such as this, a linear relationship is posited as plausible and serves as a sound first attempt at modelling the data and identifying effect. The regression model below specifies the relationship between the response variable (hotel choice preference) and the explanatory variables of frequency of posts and value of posts. Our general regression model is as follows:

$$Y_i = \beta_{k0} + \beta_k X_{ki} + \varepsilon_{ki}$$

where:

- Y is the response variable,
- Xs are the explanatory variables, and
- β s the regression coefficient that measures the effect of explanatory variables on response variables.

Our research model hypothesizes that frequency and value of firm participation in consumer communities should positively affect intentions to stay with hotel brand over the others. In addition, we added several dummy variables to test for differences in choice preference based on primary type of travel (business or leisure), reward program enrolment and

length of participation in FlyerTalk forums. (Dummy variables allow effects of categorical variables to be isolated and compared against individual categories to identify differences or the lack thereof.) Thus our measurement regression model is:

$$\begin{aligned} \text{ChoicePreference} = & \beta_0 + \beta_1 \text{Frequency} + \beta_2 \text{Value} + \beta_3 \text{DummyBusiness} \\ & + \beta_4 \text{DummyMarriott} + \beta_5 \text{DummyHilton} \\ & + \beta_6 \text{DummyYear0to1} + \beta_7 \text{DummyYear1to3} \\ & + \beta_8 \text{Dummy3to5} + \varepsilon_0 \end{aligned}$$

In line with H1 and H2 we expect positive and significant coefficients β_1 and β_2 indicating positive relationships between post frequency and post value on forum members' choice preference. Ceteris paribus, a significant and positive β_3 would indicate that the same level of hotel participation in forums more greatly influences business travellers than leisure travellers. Likewise, significant and positive β_4 , and β_5 would indicate that, holding hotel participation constant, Marriott and Hilton customers are more greatly influenced than Starwood customers in their likelihood to stay at a hotel of that brand. Significant and positive β_6 , β_7 , β_8 would indicate that people who are relatively newer to the forums are influenced more than forum veterans by hotel participation in forums.

As may be seen in the regression analysis results presented in Table 3 below, there is a positive and highly significant ($p = 0.00002$) relationship between post frequency and choice preference, offering strong support for H1. The relationship between post value and choice preference is also positive and significant ($p = 0.04279$) providing support for H2. In other words, increasing the frequency and value of firm participation in consumer communities increases forum members' intentions to stay at a hotel of that brand rather than the others, but post frequency appears to have a greater impact than does post value. Specifically, if there is a unit increase in the perception of post frequency, the preference for the hotel choice goes up by 0.72 for the base model.

Table 3 Regression results

Regression results			
Variable	Coefficient	t-Stat	p-value
Intercept	0.2481	0.5287	0.59788
Post frequency	0.4748	4.3855	0.00002***
Post value	0.2317	2.0458	0.04279**
DummyBusinessTravel	-0.1567	-0.9633	0.33715
DummyHilton	0.2457	1.0564	0.29271
DummyMarriott	0.3394	1.3908	0.16666
DummyYear0to1	0.2208	0.7889	0.43173
DummyYear1to3	0.1823	0.9254	0.35643
DummyYear3to5	0.0734	0.3139	0.75407
F-Statistics		8.32***	
R ²		0.3387	

*** indicates significance at $p < 0.01$, ** indicates significance at $p < 0.05$;

* indicates significance at $p < 0.10$

Travel type, reward program membership, and length of members' forum participation do not significantly influence forum members' choice preference. These results are profound as they suggest that, controlling for other potentially influencing factors, more frequent and valuable posts by firms to consumer communities leads to an increase in consumers' purchasing intentions.

To ensure that our regression coefficients were not inflated and poorly estimated, we tested for multicollinearity by removing, in turn, the component of each factor explained by the other and regressing using the residuals instead and this effected no change in the results. The largest Variance Inflation Factor was found to be 2.38, well below the traditional upper limit of 10 (Neter et al. 1996). Thus we conclude the issue to be mild at worst.

Responses to our survey reveal significant differences in members' perceptions of hotel representatives' post frequency and value. Our results also indicate firm representatives' post frequency and post value positively impact members' choice preference (H1 and H2 supported). These findings are consistent across differences in members' length of forum involvement, preferred rewards program, and travel type. An important and highly interesting finding is that firms may have the potential for directly and positively influencing purchasing intention by increasing the frequency of their participation in consumer communities. Advertising models have long recognized the value of frequency in keeping existing customers and attracting new buyers. In advertising, frequency is often the repetition of a single message, but in consumer communities many messages are shared. There is one constant element in the Starwood Lurker's posts however—his large and highly visible signature line that includes the text “Starwood Hotels and Resorts” and its reward program, “Starwood

Preferred Guest.” Perhaps then the Lurker's impact on members' choice preference is due, at least in part, to the repeated brand advertising his frequent posts provide. But can posting too frequently exert a negative influence on purchasing intention? Godfrey et al. (2011) found that too many communication attempts by firms seeking to strengthen relationships with consumers can backfire and drive existing customers away. We did not find evidence of this effect from the Lurker's highly active participation in FlyerTalk forums, perhaps because the quantity of firm/customer contact is controlled by forum members in their choice to visit the particular FlyerTalk forums where the Lurker is active.

Members' perception of post value positively influencing choice preferences is also an important finding. Certainly creating value is part of any effort to attract and maintain customers. Prior research suggests that perceived customer service quality positively influences purchasing intention both directly and indirectly through its impact on customer satisfaction (Zeithaml et al. 1996; Cronin et al. 2000). But what is it about the Lurker's posts that FlyerTalk members perceive as more valuable than the posts of the other hotels' representatives? One explanation for this difference may be that forum members perceive the Lurker's customer service to be of higher quality than that of the other hotels' representatives. Much of the Starwood Lurker's forum activities are direct interactions with individual customers to address their concerns, correct mistakes made by other Starwood employees and maximize customer satisfaction. These efforts stand in contrast to Hilton and Marriott representatives' forum activities, which focus primarily on promotional announcements. The Lurker's high quality customer service may lead to perceptions of high value in his posts and this is clearly more challenging for a firm to manipulate than frequency alone.

Pavlou and Dimoka's 2006 study of how customers' feedback on online sellers' credibility and benevolence assists buyers in distinguishing between sellers, may provide additional insights into the value differences found in our study. According to Pavlou and Dimoka, “credibility is the buyer's belief that a seller is competent and reliable, would perform a transaction effectively, and would acknowledge guarantees and promises” (2006, p. 395). It may be then that the Starwood Lurker's history of helpfulness, responsiveness and issue resolution on FlyerTalk gives him credibility with forum members and influences their perceptions of his posts' value. Pavlou and Dimoka define benevolence in online markets as “... the buyers' belief that a seller has beneficial motives, is genuinely concerned about the buyer's interests, and will act in a goodwill manner beyond short-term profit expectations.” (2006, p. 395). In coding actual posts made by the three hotels' representatives, we found evidence of the Lurker's sincere attempts to help forum members, including instances when he sided with members in their disputes with his own firm. We did not find participation of this nature by

the other hotel reward program representatives. Thus, another possible explanation of differences in perceived value of posts is due to respondents' perception that the Lurker is more benevolent than are the other hotel representatives. However, given the contextual differences between our study and Pavlou and Dimoka's (2006) which examined customers' text reviews of online sellers, more research is needed to determine how credibility and benevolence influence consumers' perceptions of value in firms' forum posts.

Conclusion

While these findings contribute to both the literature and to practice, our study is not without its limitations and continued work is needed to gain a fuller understanding of the phenomena at play in this context. For example, the generalizability of our study is limited by our focus on the forum participation of three specific hotels in a single consumer community. Further research might also explore the boundaries of generalizability to other service industries and possibly beyond. It may be that there are fundamental characteristics at work in our context that could be expected to lead to similar results in similar contexts. For example, FlyerTalk's success arises, arguably, from the appetite for on-going purchase support and advice over extended periods of multiple purchases that are both complex and expensive enough to warrant consumer's investment in time—an understandable consequence of the Internet having supplanted traditional travel agent services. Other travel industry segments, such as airline booking, can be characterized similarly, making them good candidates to explore for generalization while shopping for health care is probably not, despite the complexity and need for advice, since the purchase may be too infrequent to warrant the kind of long-term, on-going relationships we see in FlyerTalk.

Lastly, and necessarily, due to the preliminary nature of the study and the data collection pragmatics of the research context, perceived post value was conceptualized as a unitary comprehensive abstraction, left to subjects' individual interpretation. This is admittedly limiting but there is precedent in the literature as reported above and, to the study's credit, the validity of this conceptualization was confirmed by the finding of its significance as a determining factor, but there are undoubtedly constituent sub-factors yet to be identified and explored. Subsequent research that "drills down" into the constituent sub-factors of perceived post value would yield a more detailed, finer-grained understanding to better guide firms and their agents engaging in online consumer communities.

Furthermore, additional insights might be gained both by collecting larger data sets and/or applying alternative analysis techniques. Although our sample included a well balanced number of respondents from each hotel reward program and,

based upon demographic characteristics, appeared to represent a broad range of FlyerTalk members, future studies with larger sample sizes may be able to tease out a more refined understanding of the relationships studied here. And content analysis of the postings themselves is a promising avenue to gain deeper insight into the essential nature of posting value, particularly given the advent of automated text mining, sentiment analysis, etc. Firms seeking to maximize the effectiveness of their engagement in these communities would certainly benefit from understanding what are the nuances in writing style, tone, etc. that contribute favourably to consumers' value perceptions. These are the kinds of questions that could be addressed by applying content analysis tools such as Netnography (Kozinets 1998), a branch of ethnography focused on Internet-based communication, to study entire interaction threads, including the consumer responses, rather than just the postings themselves, and the results could be extraordinarily valuable.

Over the last decade social networking sites have altered the ways in which we communicate with others and offered new opportunities for communication between firms and their customers. In this study our results illustrate how one firm, Starwood Hotels and Resorts, engages members of the online consumer community FlyerTalk to positively influence consumers' purchasing intention more effectively than its competitors. There are several differences between Starwood's approach to forum participation and approaches taken by Marriott and Hilton and these may provide valuable insights for businesses. First, Starwood's representative, the Starwood Lurker, is much more active on FlyerTalk than the other two hotels' representatives, posting 22 times for each post by the Marriott Concierge and 55 times for each post made by the HHonorsRepresentative. Second, the Starwood Lurker leverages FlyerTalk to offer individualized customer service to members, increase brand awareness and manage brand image while Marriott and Hilton representatives use FlyerTalk primarily as a means to broadcast rewards program promotions. Third, the Starwood Lurker's full time job is to monitor FlyerTalk and to engage its members in many different forums on the site. The Marriott Concierge and HHonorsRepresentatives are teams of employees who participate solely in their own forums and only as often as their other commitments permit. Lastly, the Starwood Lurker also participates in FlyerTalk as a fellow member and frequent traveller independent of his role as a representative of Starwood Hotels and Resorts so the members may feel greater camaraderie with him. We did not find evidence of such personal participation by the Marriott Concierge or the HHonorsRepresentative.

Many questions regarding firm participation in consumer communities remain. What additional aspects of firm participation most greatly influence consumer community members' purchasing intentions? Do the revenues generated by increased purchasing intention outweigh forum representatives'

employment costs? Can the Starwood Lurker's successes be replicated by firm representatives in other industries? Only further research can find the answers, but the implication of this study for today's businesses is clear: firms are generating value by participating in online consumer communities and engaging members in meaningful ways. Companies can either capitalize on the opportunities these and other social networking sites provide or surrender the potential rewards to their competitors.

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